SIMATS ENGINEERING
ASSESSMENT TOOL 1
COURSE NAME: Artificial Intelligence COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

Register Number	**192524358**
Name	**Abdul Razak Shaik**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks: 50**

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

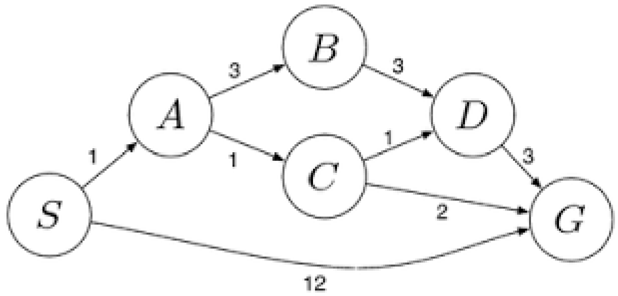
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost

path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

*I ____**AbdulRazak Shaik(192524358)**____________ (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:AbdulRazak Shaik

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10

Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

CSA1709 -Artificial Intelligence

CO -1 ASSESSMENT TOOL 1

Question 1: Water Jug Problem

Problem

Two jugs are available:

- Jug A = 4 gallons
- Jug B = 3 gallons

The Main objective is to obtain exactly **2 gallons in the 4-gallon jug**.

Initial State

(0,0)

Both jugs are empty.

Goal State (2,0)

The 4-gallon jug contains exactly 2 gallons.

Operators

- Fill either jug completely.
- Empty either jug.
- Pour water from one jug to another until one becomes empty or the other becomes full.

Algorithm: Water Jug Problem Using State Space Search

Step 1: Start.

Step 2: Initialize both jugs as empty.

State = (0,0)

Step 3: Fill the 3-gallon jug.

Step 4: Pour water from the 3-gallon jug into the 4-gallon jug.

Step 5: Fill the 3-gallon jug again.

Step 6: Pour water into the 4-gallon jug until it becomes full.

Step 7: Empty the 4-gallon jug.

Step 8: Transfer the remaining 2 gallons from the

3-gallon jug to the 4-gallon jug.

Step 9: Check whether the 4-gallon jug contains

exactly 2 gallons.

Step 10: If yes, display Goal State.

Step 11: Stop.

Solution Table

Step	Action	State (4-Gallon, 3-Gallon)
1	Fill 3-gallon jug	(0,3)
2	Pour into 4-gallon jug	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4-gallon jug until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons into 4-gallon jug	(2,0)

State Space Diagram

```
        Start
        (0,0)
          |
     Fill 3-Gallon Jug
          |
        (0,3)
          |
  Pour into 4-Gallon Jug
          |
        (3,0)
          |
```

Fill 3-Gallon Jug
|
(3,3)
|
Pour into 4-Gallon Jug
|
(4,2)
|
Empty 4-Gallon Jug
|
(0,2)
|
Pour into 4-Gallon Jug
|
(2,0)

GOAL STATE

Question 2: Mars Rover Intelligent Agent

i) Percepts

The Mars Rover receives information through:

- Camera images
- Rock and soil samples
- Temperature readings
- Atmospheric pressure
- Radiation level
- Wheel movement sensors
- GPS/Navigation sensors
- Battery level

ii) Operating Environment

Property	Description
Observable	Partially Observable
Agents	Single Agent
Deterministic	Mostly Stochastic
Sequential	Yes
Dynamic	Yes
Continuous	Yes

iii) Actions

The Mars Rover can:

- Move forward/backward
- Turn left/right
- Capture images
- Drill rocks
- Collect soil samples
- Analyze samples
- Send data to Earth
- Avoid obstacles
- Recharge using solar panels

iv) Performance Measure

The rover is evaluated based on:

- Number of successful samples collected
- Accuracy of scientific observations
- Safe navigation
- Battery efficiency
- Communication success
- Mission completion time
- Obstacle avoidance

v) Suitable Agent Architecture

Model-Based Goal-Based Agent

Reason:

- Maintains an internal model of Mars.
- Plans routes to scientific targets.
- Handles uncertain environments.
- Makes intelligent decisions independently
- Updates knowledge continuously.

A **Model-Based Goal-Based Agent** is the most suitable architecture because Mars is only partially observable and highly dynamic.

Question 3: 8-Queens Problem

Problem Formulation

Place **8 queens** on an **8 × 8 chessboard** so that no two queens attack each other.

Components

Initial State

Empty chessboard.

State Space

Every possible arrangement of queens on the board.

Operators

Place one queen in a safe position.

Goal Test

No two queens should attack each other.

That means:

- Same row
- Same column
- Same diagonal

Path Cost

Every queen placement has equal cost.

Search Strategy

Backtracking Search (Depth First Search)

Solution Representation

Example solution:

Column	Row
1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

Search Tree

```
Start
  |
Q1
  |
Q2
  |
Q3
  |
Q4
  |
Q5
  |
Q6
  |
Q7
  |
Q8
  |
Goal
```

Goal State Example:

Row\Coloumn	C1	C2	C3	C4	C5	C6	C7	C8
R1	Q	–	–	–	–	–	–	–
R2	–	–	–	–	–	–	Q	–
R3	–	–	–	–	Q	–	–	–
R4	–	–	–	–	–	–	–	Q
R5	–	Q	–	–	–	–	–	–
R6	–	–	–	Q	–	–	–	–
R7	–	–	–	–	–	Q	–	–
R8	–	–	Q	–	–	–	–	–

Backtracking efficiently eliminates invalid placements and finds one of the valid solutions.

Question 4: OLA Cab Booking

Problem-Solving Agent

A **Goal-Based Agent** is most suitable.

Reason

The customer wants to reach a destination with preferred cab type.

The agent searches available cabs and selects one satisfying the goal.

Inputs

- Pickup location
- Destination
- Cab type
- Payment mode

Output

- Confirmed booking
- Driver details
- Estimated fare
- ETA

Pseudocode:

```
START

Input Pickup Location
Input Destination
Input Cab Type

Search Available Cabs

IF Cab Available THEN
       Display Fare
       Confirm Booking
       Assign Driver
```

```
        Show ETA
ELSE
        Display "No Cab Available"
ENDIF

END
```

Flow

```
Start
   |
Enter Pickup
   |
Enter Destination
   |
Select Cab
   |
Search Cab
   |
Cab Available?
 /       \
Yes       No
 |         |
Book     Show Error
 |
Driver Assigned
 |
End
```

A Goal-Based Agent helps achieve the customer's objective of reaching the destination efficiently.

Question 5: Uniform Cost Search (UCS)

Uniform Cost Search

Uniform Cost Search always expands the node with the **lowest cumulative path cost**.

Algorithm

1. Start from source node **S**.
2. Insert S into the priority queue.
3. Remove the node having minimum cost.
4. Expand all neighbouring nodes.
5. Update cumulative costs.
6. Continue until destination G is reached.

Example

Suppose the graph is

```
        2
   S --------- A
   |           |
 4 |           |3
   |           |
   B --------- C
    \         /
     5       2
       \   /
        G
```

UCS Table

Expanded Node	**Cost**
S	0
A	2
B	4
C	5
G	7

Search Tree:

```
         S(0)
       /    \
     3 /     \2
      A       B
      |      / \
     4 |   2 /  \6
      |    /    \
      D    C     G
      |   / \
     5 |  2/  \3
```

```
  | /   \
 G D     G
```

Least Cost Path

S → A → C → G

Total Cost

2 + 3 + 2 = 7

Advantages

- Finds optimal solution.
- Complete algorithm.
- Works for different edge costs.
- Suitable for route planning and logistics.

Uniform Cost Search guarantees the **minimum-cost path** from the starting warehouse to the destination warehouse.